AI IN STRUCTURAL HEALTH MONITORING FOR PREDICTIVE MAINTENANCE & SAFETY ASSURANCE

EXECUTIVE PROGRESS SUMMARY: PRIOR MDP, CURRENT IDP, DUAL-PLATFORM DEPLOYMENT & PROCUREMENT PLAN

BACSE291 -- Innovative Design Project | Fall 2026 | Vellore Institute of Technology (VIT), Vellore

Guide: Dr. Senthil Kumar N (Emp ID: 15377, SCOPE) | **Aerial (15377IDP01):** Priyam S., Ayushman K. | **Ground (15377IDP02):** Nisargkumar J. (Lead), Aryan M., Mudit G.

1. PREVIOUS MDP PROJECT: PLY FORGE POINT CLOUD PROCESSOR

• **Architecture & Independence:** Client-side WebGL 1.0 3D point cloud processor in HTML5/ES6 requiring zero external libraries (no Three.js, no Node.js backend).
• **Core Capabilities:** Binary/ASCII PLY parser (`DataView`/`ArrayBuffer`), built-in 18,000-pt Stanford Bunny demo, circle ROI filtering, 2.5D Delaunay/grid meshing, and PLY export.
• **Voxel Downsampling:** 3D spatial hash map downsampling point clouds by 60%--90% while preserving structural boundaries.
• **Direct IDP Role:** Directly embedded into Ground Station Tab 4 (Streamlit WebGL Viewer) providing zero-latency 3D facade rendering without external desktop software.

2. CURRENT IDP WORK DEVELOPED & VERIFIED

• **Aerial Sensing (15377IDP01):** Calibrated Benewake TF-Luna 850nm dToF (<5g, 0.35W, 100 Hz) on ESP32 FreeRTOS. UART sync to 0x59 0x59 with checksum verification and median filter (N=5).
• **Empirical Metrology Fit:** Universal regression model: $y = 1.0228x + 3.20\ cm$ (R^2=1.0000, RMSE=±0.02 cm across 39.0cm and 61.4cm physical benchmarks).
• **Ground Analytics (15377IDP02):** Rolling baseline tracking calculating geometric residuals (Δd); flags spalls (Δd ≥ +3.5cm) and bulges (Δd ≤ -3.5cm).
• **3D Digital Twin:** Reconstructed 9,969-point room (.PLY) isolating Almari bulge (+65cm, Orange) and Gate recess (-60cm, Green). 5-tab Streamlit dashboard active.

3. DUAL-PLATFORM STRATEGY & HARDWARE UPGRADES

• **Platform Duality:** (1) *Autonomous F450 Drone:* Elevated facade & bridge inspection; (2) *Smart 4WD Ground Car:* Indoor corridors, eliminating prop-wash suction & hand jitter.
• **Smart 4WD Car Odometry:** Dual 20-slot optical wheel encoders integrate distance at ~1.02 mm/tick; closed-loop PID maintains constant 5.0 cm/s traverse; coordinates: (X_enc, d_meas*cosθ, Z_mast).
• **YDLIDAR X2 360° LiDAR (INR 7,189.00):** 3,000 pts/s, 7 Hz planar rotation, 0.12--8.0m range, capturing full 2D cross-sectional wall slices in 142 ms.
• **DFRobot Gravity Tilt Sensor (INR 245.00):** Instant pitch/roll correction: d_true = d_meas*cos(θ), eliminating 3.1 cm artificial elongation false cavities.

4. FUTURE ROADMAP & SYSTEM INTEGRATION

• **Phase 1--2 (Weeks 1--3):** Procure Tier 1 hardware (LiDAR, tilt sensor, 4WD car kit); verify UART framing and 5V rail regulation.
• **Phase 3--4 (Weeks 4--6):** Assemble F450 airframe & propulsion; implement silicone vibration-damping deck (>18 dB attenuation); benchmark hover thrust.
• **Phase 5--6 (Weeks 7--10):** Laboratory wall scanning with Smart 4WD car (encoder odometry); tethered facade flights; blind defect detection benchmarks.
• **Phase 7 (Weeks 11--14):** End-to-end integration, Streamlit PLY Forge 3D meshing, automated compliance PDF reports, and IEEE conference paper submission.
• **Target Benchmarks:** Static distance RMSE ≤ ±1.5cm; Tilt residual ≤ 0.5cm; Defect sensitivity ≥ 95%; Car odometry drift ≤ 1.0cm / 5m.

Component Group	Key Inclusions & Specifications	Cost (INR)	Procurement Phasing
Sensors & Contingency Car	YDLIDAR X2 360° LiDAR (INR 7,189), DFRobot Tilt (INR 245), Smart 4WD Car + Encoders (INR 1,750), Buck (INR 240)	**INR 9,424.00**	**Tier 1 (Immediate Critical / Weeks 1--2)**
Airframe & Power Reservoir	F450 Frame + 4x Motors + 4x ESCs (INR 5,200), 1045 Props (INR 260), 3S 4200mAh LiPo & B6 Charger (INR 4,430)	**INR 9,890.00**	**Tier 2 (UAV Assembly / Weeks 3--4)**
Avionics, Shipping & Reserve	Pixhawk 2.4.8 Kit (INR 5,950), ESP32 (INR 380), Dampers/Wire (INR 570), Shipping (INR 400), Contingency (INR 2,000)	**INR 9,300.00**	**Tier 3 (Flight Trials / Weeks 5--6)**
TOTAL DUAL-PLATFORM BUDGET	**Complete Airborne UAV & Smart 4WD Ground Robotic Car Systems**	**INR 28,874.00**	**Ground MVP Option: INR 9,964.00**